

After completion of the preparation materials, students should be able to:

1. SNS physiology → Mechanisms → direct-acting pharmacologic effects

Predict the therapeutic effects of drugs that selectively activate specific adrenergic receptors.

2. SNS physiology → Mechanisms → indirect-acting pharmacologic effects

Predict the sympathomimetic effects of drugs that cause the release of norepinephrine from storage vesicles, block uptake by NET, or inhibit metabolism of norepinephrine.

3. SNS physiology → Mechanisms → mixed-acting pharmacologic effects

Predict the sympathomimetic effects of drugs that cause the release of norepinephrine from storage vesicles and act directly on adrenergic receptors.

4. Pharmacologic effects → therapeutic uses

Relate the specific sites of action and effects to the therapeutic uses of sympathomimetics in this presentation.

5. Actions → adverse effects and cautions

Predict the adverse effects, drug interactions, and contraindications based on the actions of the drugs influencing sympathetic nervous system functions.

Key points:

1. SNS physiology → Mechanisms → direct-acting pharmacologic effects:

The synthetic direct-acting agonists in Part 2 of this presentation bind directly to adrenergic receptors β_2 , β_3 , and α_2 , receptors. They are widely used in clinical practice.

2. SNS physiology → Mechanisms → indirect-acting and mixed-acting pharmacologic effects

Indirect-acting adrenergic agonists cross the blood-brain barrier and cause the release, inhibit the reuptake, or inhibit the degradation of biogenic amines. They potentiate the effects of endogenous epinephrine or norepinephrine but do not directly bind to or affect postsynaptic receptors.

Mixed-acting drugs act centrally and peripherally to cause the release of NE and EPI from storage vesicles and direct action on α and β adrenergic receptors.

3. Pharmacologic effects → therapeutic uses (next slide)

Direct-acting sympathomimetics have various therapeutic uses.

Indirect-acting sympathomimetics are used (and abused) for their sympathomimetic effects in the CNS.

Mixed-acting agents, ephedrine and pseudoephedrine, have limited use.

4. Actions → adverse effects and cautions

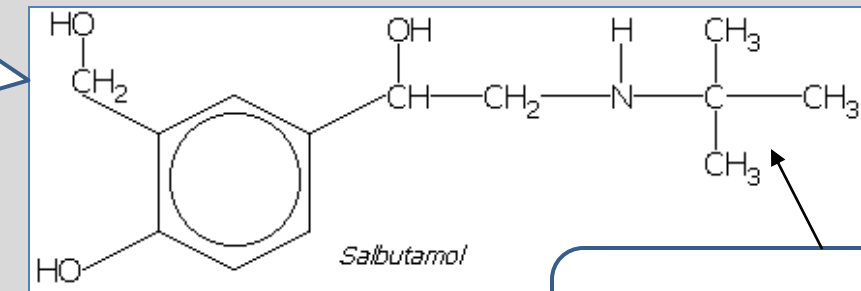
Adverse effects and cautions are typical of the sympathomimetics class.

Key points: Mechanisms → Effects → Therapeutic uses

- Beta-2 agonists induce bronchial smooth muscle relaxation and bronchodilation. They are administered primarily by oral inhalation for direct action on lung tissue in the management of asthma and chronic obstructive pulmonary disease (COPD).
- Beta-3 agonists relax urinary bladder smooth muscle for the treatment of overactive bladder.
- Centrally-acting alpha-2 agonists act by reducing sympathetic outflow from the CNS. Thus, they are useful in the management of hypertension. Clonidine is also used for reducing withdrawal symptoms related to substance abuse. Methyldopa is used primarily for management of chronic hypertension in pregnancy.
- Indirect-acting sympathomimetics, amphetamines, MAO inhibitors, and catecholamine reuptake inhibitors increase synaptic levels of norepinephrine. Therapeutic uses are attention deficit disorder and depression.
- Mixed-acting sympathomimetics (eg, ephedrine) cause the release of norepinephrine from storage vesicles and act directly on adrenergic receptors. Ephedrine is used parenterally for anesthesia-induced hypotension and orally as a bronchodilator. Pseudoephedrine is less potent and taken orally for nasal decongestion.

Selective β_2 Agonists

Resistance
to COMT



β_2 selectivity and
↓ MAO metabolism

Short-acting beta agonists (SABAs): On demand use

*Albuterol Metaproterenol (less selective than the others)

Levalbuterol Terbutaline

Long-acting beta agonists (LABAs): (twice daily dosing)

*Salmeterol *Formoterol Arformoterol

Ultra-LABAs: (once daily dosing)

Vilanterol, Indacaterol, Olodaterol

Beta 2 Agonists Class Pharmacokinetic Properties

	SABAs	LABAs
A	Oral systemic; Oral inhalation; I.V.	Oral inhalation
F	Some absorption from the lungs	
D	Low plasma protein binding	
M	Hepatic conjugation inactive metabolites	Hepatic CYP metabolism
E	Renal excretion unchanged and metabolites	Renal and fecal excretion of metabolites
t _{1/2}	4-5 hours	5 to 50 hours (product specific)
Onset, peak	~10 minutes	Formoterol ≤3 min – rapid-acting LABA Other LABAs 15-19 minutes
Duration	4-6 hours	~12 hours Ultra-LABA: ~24 hours

Formoterol
takeaway

F: bioavailability; SABA: short-acting beta agonist; LABA: long-acting beta agonist

Routes of Administration: Clinical Considerations

Aerosol:

Delivers drug to distal airways

- Small doses activate β_2 receptors in bronchial smooth muscle.
- SABAs (rapid-acting versions) provide rapid bronchodilation.
- LABAs (long-acting / ultra-long-acting versions) have a slower onset and a long-lasting effect.
- ~10% of orally inhaled dose enters the lungs.
- The remainder is swallowed and ultimately absorbed.

Oral β_2 AR agonists:

- No advantage over the inhaled forms
- Less effects and associated with increased incidence of side effects
- Alternative for patients in whom aerosol delivery of drug causes severe exacerbation of bronchospasm from local irritation

Selective β_2 Agonists: Mechanisms and Therapeutic Effects

- Agonists of β_2 AR → **induce relaxation of bronchial and uterine smooth muscle**

Long-acting versions are highly lipophilic + high affinity for β_2 ARs and they are less susceptible to metabolism by COMT and MAO compared to the short-acting versions.

Effects

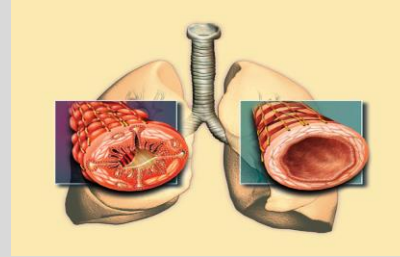
1. **Bronchodilation → relieve bronchospasm**
2. May suppress release of leukotrienes and histamine from mast cells in lung tissue → prevent symptoms
3. May enhance mucociliary function, ↓ microvascular permeability, and possibly inhibit phospholipase A_2

Note: Glucocorticoids and leukotriene receptor antagonists have better anti-inflammatory effects in asthma.

4. **Tocolytic action: Slow uterine contractions → suppress labor**

Selective β_2 Agonists Therapeutic Uses

Asthma and COPD



Rapid-acting

albuterol, levalbuterol, formoterol (a LABA)

- Relief of acute bronchospasm
- Exercise-induced bronchospasm, prevention as needed

Long-acting

- Prophylaxis of bronchospasm (with inhaled GC)
- Symptomatic nocturnal asthma
- Exercise-induced bronchospasm, prevention

Tocolytic effects - terbutaline

(uterine relaxation to delay labor)

- Activate cAMP, PKC $\rightarrow\rightarrow\rightarrow$ inactivates myosin light-chain kinase $\rightarrow$ relaxation of myometrium
- Delay birth x48 hours to administer a glucocorticoid to the mother to promote fetal lung maturation and transport mother to hospital with a neonatal unit
- AEs: tachycardia; hypotension; pulmonary edema



Selective β_2 Agonists Adverse Effects

Tremor:

Skeletal muscle β_2 activation
Tolerance may develop

Tachycardia:

Stimulation of cardiac β_2 ARs and
reflex tachycardia may be involved
(extent is uncertain)

Metabolic effects

with parenteral administration

hypokalemia (K^+ uptake by skeletal
muscle)

hyperglycemia (hepatic
gluconeogenesis, glycogenolysis)

↑ plasma free fatty acids (lipolysis)

Restlessness; apprehension; anxiety

Occurs with systemic administration
(oral, parenteral)

Selective β_3 Agonist: Mirabegron

PK	Oral, CYP2D6 inhibitor, $t_{1/2}$ ~50 hours, onset within 8 weeks
Mechanism Figure on previous slide	Activation of β_3 in urinary bladder → relaxation of detrusor smooth muscle during storage phase → enhanced bladder capacity
Therapeutic use	Treatment of overactive bladder with symptoms of urinary frequency, urgency, or urge incontinence May be used in combination with an antimuscarinic drug. <i>Mirabegron lacks cholinergic side effects.</i>

Recall: M_3 receptor antagonists are also used for management of overactive bladder. They inhibit the actions of ACh on bladder smooth muscle → ↓ bladder spasms, ↑ bladder capacity, and decrease urgency and frequency (onset of action 30-60 min).

Adverse effects Dose-dependent	↑BP, monitor BP; QT interval prolongation (low risk)
Cautions	Avoid in patients with severe hypertension or at significant risk of QT interval prolongation (history, medications)
Drug interactions	CYP2D6-mediated Mirabegron may increase serum concentrations of 2D6 substrates.

Check your knowledge

- What is the mechanistic effect of SABAs and LABAs (includes “ultra”-LABAs)?
- What are the therapeutic goals of SABAs and LABAs?
- What are common side effects of beta-2 agonists?
- Why is inhalation the preferred route of administration for beta-2 agonists?

- What is the function of beta-2 receptors on uterine smooth muscle during pregnancy.
- What is the clinical rationale for systemic administration of a beta-2 agonist to a pregnant person between 24 weeks and 34 weeks gestation as tocolytic therapy?
- What is the current clinical use of beta-3 agonists?
- What is the mechanism of action for this therapeutic use?
- What is the name of beta-3 agonist you should be able to discuss for this lecture?



Methyldopa Therapeutic Uses and Adverse Effects

Therapeutic use

- **Hypertension**, moderate-to-severe
- **Pregnancy hypertension** treatment
 - Methyldopa lacks fetal adverse effects,
 - maintains uterine blood flow,
 - and does not reduce maternal cardiac output or blood flow to the kidneys.

Adverse effects:

dry mouth; sedation; low energy; depression; headache; orthostatic hypotension; ejaculation disorder; impotence; reduced libido

Endocrine effects:

Parkinsonian symptoms; hyperprolactinemia / gynecomastia and galactorrhea

Uncommon but potentially serious

Hepatitis, hemolytic anemia; aplastic anemia, and drug fever

Check your knowledge

- How does activation of alpha-2 receptors lead to reduction in blood pressure?
- What is the proposed mechanism as to how activation of alpha-2 receptors provides pain relief?
- What is a proposed mechanism as to how activation of alpha-2 receptors is useful for the management of ADHD?

Check your knowledge

- How are clonidine and methyldopa the same as each other and different from each other?

Commonly used drugs such as antidepressants, amphetamines, and cocaine target monoamine (norepinephrine, dopamine, and serotonin) transporters with different potencies.

- Panel A: Mechanism of NE reuptake into the noradrenergic neuron via the NET, where a proportion is sequestered in presynaptic vesicles through VMAT.

(This mechanism applies to all biogenic amines.)

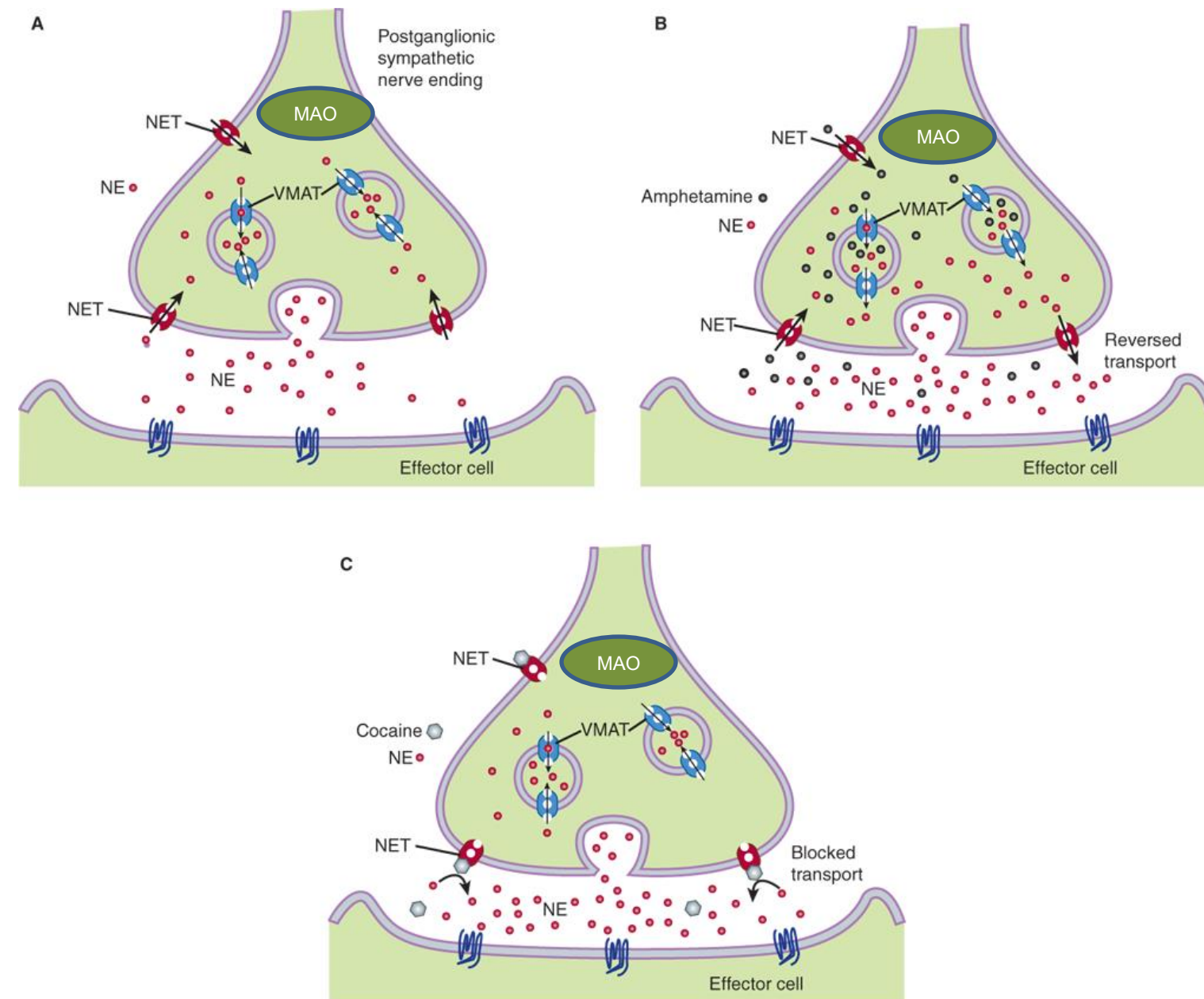
- Panel B: Amphetamine (dark dots) causes release of biogenic amines from storage sites, blocks NET, and causes reverse transport of neurotransmitter.

NE accumulates in the synapse and beyond the synaptic cleft into the extraneuronal space.

- Panel C: Cocaine blocks the NET (and DAT) inhibiting uptake of the neurotransmitters.

- Inhibition of monoamine oxidase (MAO) increases synaptic stores of NE (and other biogenic amines), which enhances neurotransmitter effects.

MAO



Source: Bertram G. Katzung, Todd W. Vanderah:
Basic & Clinical Pharmacology, Fifteenth Edition
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NET: norepinephrine reuptake transporter
VMAT: vesicular monoamine transporter

- Beta-2 receptor agonists are inhaled bronchodilators in the management of asthma and COPD. Short-acting agents (SABAs) provide rapid relief of bronchospasm. Long-acting versions (LABAs) are for chronic maintenance to prevent attacks.
- Centrally acting alpha-2 receptor agonists, such as clonidine and methyldopa, are sympatholytics – they block sympathetic outflow from the CNS for the treatment of hypertension. Clonidine has several other uses as well.
- Indirect-acting sympathomimetics are amphetamine-like drugs, MAO and COMT inhibitors, and norepinephrine reuptake (NET) inhibitors.
- Amphetamines promote the release of catecholamines from storage vesicles (predominant effect), primarily norepinephrine and dopamine. Amphetamines may also block reuptake of catecholamines by competitive inhibition at the transporters. These drugs are used in the management of attention deficit disorder. They readily enter the CNS and have marked stimulant effects on mood and alertness and depressant effect on appetite. Amphetamines have a high potential for abuse.

- Cocaine blocks catecholamine transporters. Its major action is inhibition of norepinephrine and dopamine reuptake into neurons in the pleasure centers of the brain. It is a CNS stimulant that produces intense amphetamine-like effects. Cocaine also has local anesthetic actions.
- The serotonin-norepinephrine reuptake inhibitors (SNRIs) are used in depression and anxiety disorders, for chronic pain, and for the pain of fibromyalgia.
- MAO inhibitors are used for treatment-resistant major depressive disorder. They cause severe drug-drug and drug-food interactions by inhibiting tyramine metabolism by MAO-A, which can result in severe hypertension.
- COMT inhibitors prolong the effects of levodopa by preventing peripheral metabolism for the symptomatic management of Parkinson disease.
- Mixed acting agents, ephedrine and pseudoephedrine (less potent isomer of ephedrine), cause the release of NE (indirect action) and have agonist actions at α and β receptors. Ephedrine is used for anesthesia-induced hypotension and orally as a bronchodilator. Pseudoephedrine is a nasal decongestant.

- What is the mechanistic effect of SABAs and LABAs (includes “ultra”-LABAs)?
- Beta-2 agonists relax bronchial smooth muscle and relieve bronchospasm.
- What are the therapeutic goals of SABAs and LABAs?
- SABA: As needed (PRN) use for relief of acute bronchospasm (“rescue inhaler”) and prevention of exercise-induced bronchospasm.
- LABA: Chronic use for prevention of bronchospasm
- What are common side effects of beta-2 agonists?
- Tremor, tachycardia, nervousness. Side effects are reduced by inhalation compared to oral administration.
- Parenteral beta-2 agonist also causes metabolic effects, hypokalemia, hyperglycemia, and increased serum free fatty acids.
- Why is inhalation the preferred route of administration for beta-2 agonists?
- A small dose delivers a high concentration of drug to the distal airways.
- Rapid relief of bronchospasm. Generally slower with LABAs but still faster compared to oral administration.
- LABAs and ultra-LABAs have a long duration of action.
- Reduced adverse effects compared to oral or parenteral administration.
- Metered dose inhalers are portable and easy to use.

Check your knowledge

- What is the function of beta-2 receptors on uterine smooth muscle during pregnancy.
- Beta-2 agonists are highly expressed on uterine smooth muscle throughout pregnancy, particularly in the third trimester. Beta-2 receptors are important for preventing premature contractions.
- What is the clinical rationale for systemic administration of a beta-2 agonist to a pregnant person between 24 weeks and 34 weeks gestation as tocolytic therapy?
- Systemic administration of a beta-2 agonist (typically terbutaline) along with a long-acting glucocorticoid for 48 hours can delay labor to enhance fetal lung development (the purpose of glucocorticoid administration to the mother).
- What is the current clinical use of beta-3 agonists?
- Treatment of overactive bladder with symptoms of urinary frequency, urgency, or urge incontinence. It may be used in combination with a muscarinic antagonist.
- What is the mechanism of action for this therapeutic use?
- Activation of β_3 in urinary bladder → relaxation of detrusor smooth muscle during storage phase → enhanced bladder capacity
- What is the name of beta-3 agonist you should be able to discuss for this lecture?
- Mirabegron

Check your knowledge

- How does activation of alpha-2 receptors lead to reduction in blood pressure?
- Activation of central postsynaptic and presynaptic alpha-2 GiPCRs activates inhibitory neurons in the brainstem, which reduces sympathetic outflow, which leads to vasodilation and decreased vascular resistance, decreased heart rate, and decreased BP.
- What is the proposed mechanism as to how activation of alpha-2 receptors provides pain relief?
- Epidural clonidine activation of spinal postsynaptic and presynaptic alpha-2 receptors prevents pain signal transmission to the brain.
- What is a proposed mechanism as to how activation of alpha-2 receptors is useful for the management of ADHD?
- Alpha-2 agonists, such as guanfacine and clonidine, may regulate subcortical activity in the prefrontal cortex, the area of the brain responsible for emotions, attentions, and behaviors and causes reduced hyperactivity, impulsiveness, and distractibility.

Check your knowledge

- How are clonidine and methyldopa the same as each other and different from each other?

Centrally-acting alpha-2 receptor agonist activate pre- and postsynaptic alpha-2 receptors Decrease central sympathetic outflow, heart rate and cardiac output; decrease peripheral resistance; decrease blood pressure	
Clonidine	Methyldopa
Clonidine acts directly on central and peripheral alpha-2 receptors. Activation of central imidazoline receptors may contribute to blood pressure lowering effects.	α -Methyldopa is converted in adrenergic presynaptic vesicles to α -methyl norepinephrine. It replaces NE and is released into the synapse as a false transmitter.
Bradycardia, dry mouth, and sedation are the most common adverse effects. Orthostatic hypotension may occur. Clonidine is an add-on, not first-line, antihypertensive drug.	Side effects include sedation, low energy feeling, depression. Rarely occurring effects include hemolytic anemia. Parkinsonian effects, gynecomastia, and hyperprolactinemia may occur with long term use. Methyldopa is used for blood pressure management in pregnant persons.
When discontinuing therapy, the dose of an alpha-2 agonist should be gradually tapered to prevent rebound hypertension.	

Check your knowledge

- What are the three mechanistic categories (subclasses) of sympathomimetic drugs?
- Direct-acting – selective and nonselective
- Indirect-acting
- Mixed-acting
- What are the definitions of each subclass?
- Direct-acting sympathomimetics may or many not be catecholamines. They bind directly to adrenergic receptors and activate them, producing effects like those of the natural catecholamines, epinephrine and norepinephrine.
- Indirect-acting sympathomimetics increase the concentration of norepinephrine in the synaptic cleft.
- Mixed-acting sympathomimetics both increase concentrations of norepinephrine in the synaptic cleft and directly bind and activate adrenergic receptors.